

Standard Operating Procedure **10 Ton Shop Press**

SOP: Enerpac IPH1234 10-Ton Hydraulic H-Frame Press

The Enerpac IPH1234 is a manual H-frame press designed for bending, straightening, and pressing components such as bearings and bushings. It utilizes an RR1010 double-acting cylinder and a P84 two-speed hand pump to deliver up to **10 tons (20,000 lbs)** of force. This machine is a precision instrument; its safety relies on perfectly centered loads and the management of high stored energy within workpieces.

Potential Hazards

- **Mechanical (High Pressure):** Stored energy can cause parts to "pop" or fly out of the press with deadly force if the load is not perfectly centered.
- **Crush Points:** Extreme pinch points exist between the ram and the workpiece, and between the workpiece and the bed plates.
- **Structural Failure:** Overloading the frame or using improper pins/mandrels can cause the press to deform or collapse.
- **Pneumatic/Hydraulic:** High-pressure air lines and hydraulic fluid leaks (pinhole leaks can penetrate skin).

Personal Protective Equipment (PPE)

PPE Item	Requirement
Eye Protection	**Safety Goggles (ANSI Z87.1) + Face Shield:** Mandatory. Under high pressure, brittle metals or bearings can shatter and eject at ballistic velocities.
Foot Protection	**Steel-Toed Boots:** Mandatory. Heavy arbor plates (V-blocks) and workpieces present significant crush hazards if dropped.
Hand Protection	**Heavy-Duty Work Gloves:** Recommended for handling heavy steel stock, though fingers must stay clear of the ram's travel path.
Apparel	Secure loose clothing and lanyards. Remove all jewelry, particularly rings and watches, which can be snagged during setup.



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Before Use

- **Hardware Inspection:** Verify that the bed support pins are fully seated through both sides of the H-frame.
- **Hydraulic Integrity:** Inspect hoses for kinks, bulges, or leaks. Ensure the GA2 Gauge Adapter and GF10P gauge are securely attached and legible.
- **Bed Leveling:** Ensure the vertical "daylight" is adjusted correctly for the workpiece. On this 10-ton model, bed adjustment is manual; ensure the bed is level before loading.
- **Cylinder Positioning:** The roller head design allows for side-to-side movement. Ensure the cylinder is locked in position directly over the intended press point.
- **Workholding:** Use only rated V-blocks or arbor plates. Ensure the workpiece is stable and flat against the support.

During Use

- **The "Center Rule":** Align the workpiece so the force is applied directly through its center of gravity. Offset loading causes the ram to kick, which can eject the part or bend the frame.
- **Initial Contact:** Use the P84 hand pump to slowly lower the RR1010 cylinder until it makes contact. Re-verify alignment before applying full pressure.
- **Gauge Monitoring:** Watch the GF10P pressure gauge. If the pressure rises without the ram moving, the part is seized or the press is reaching its mechanical limit. **Stop immediately.**
- **The "Line of Fire":** Stand to the side of the press while pumping, never directly in front of the open frame, to avoid potential projectiles.
- **Pumping Logic:** Use the P84's two-speed feature for rapid advance, but switch to high-pressure mode for the actual press operation.

After Use

- **Pressure Release:** Slowly turn the release valve on the P84 pump. Allow the cylinder to retract fully before attempting to remove the workpiece.
- **Storage:** Lock the pump handle in the down position for transport/storage.



- **Cleanup:** Return arbor plates and mandrels to their designated storage. Wipe down any hydraulic fluid residue.
- **Housekeeping:** Sweep the area around the press. Ensure the H-frame is clear of all tools

Emergency Equipment and Procedures

- **Hydraulic Leak:** If a leak is suspected, **do not** use your hand to find it. Use a piece of cardboard. If skin is penetrated by fluid, seek medical attention immediately.
- **Structural Noise:** If the frame "creaks" or deforms under load, release pressure immediately and re-evaluate the setup.
- **Reporting:** Report all "near-misses," leaking seals, or bent hardware to the Laboratory Project Supervisor immediately.